

MOLECULEIQ RESEARCH REPORT METFORMIN

Overall Score: 75.8/100
Confidence: 100%
Date: 2026-07-23

1. EXECUTIVE SUMMARY

Strategic Intelligence Summary for Metformin:

Metformin presents an overall Opportunity Score of 75.8/100 with a Data Confidence Score of 100.0/100. Analysis across available domains indicates 10 total clinical trial(s), 128,952 scientific publication(s), and an FTO assessment of 'At Risk — 3 active patent constraint(s)'. Target molecule research was conducted across 4 available domain dataset(s) (clinical, literature, market, patent).

2. COMMERCIAL OPPORTUNITY & SCORES

Commercial Evaluation:

Global market size is estimated at \$5,120.8 Million USD with an annual CAGR growth rate of 6.2%. Market presence spans 4 key geographic region(s) (Global, United States, Europe, Asia Pacific). Market Opportunity Sub-Score: 90.0/100.

Scoring Category	Score (0-100)	Weight	Strategic Assessment
Market Feasibility	90.0	30%	Global volume & CAGR growth
Clinical Momentum	85.0	25%	Active & completed trial validation
Patent Landscape / FTO	30.0	25%	FTO status & patent constraint risk
Scientific Research	100.0	20%	Publication volume & citation impact
OVERALL OPPORTUNITY SCORE	75.8	100%	Composite weighted opportunity score
DATA CONFIDENCE SIGNAL	100.0	Signal	Data completeness across 4 domains

3. CLINICAL DEVELOPMENT LANDSCAPE

Clinical Trial Landscape:

Identified 10 total clinical trial record(s) on ClinicalTrials.gov. Currently 2 trial(s) are actively recruiting or in progress, while 6 trial(s) have reached completion. Clinical Development Sub-Score: 85.0/100.

NCT ID	Study Title	Status	Phases	Lead Sponsor
NCT01083212	To Evaluate the Effect of Gemfibrozil on the Pharmacokinetic...	COMPLETED	PHASE1	AstraZeneca

NCT07109700	A Study of HDM1005 in Participants With T2DM Not Controlled ...	RECRUITING	PHASE2	Hangzhou Zhongmei Huadong Pharmaceutical Co., Ltd.
NCT01217307	Metformin to Reduce Heart Failure After Myocardial Infarctio...	COMPLETED	PHASE2, PHASE3	University Medical Center Groningen
NCT01593137	A Long-term Trial to Compare the Effects of Liraglutide and ...	WITHDRAWN	PHASE4	Fundación Fernández-Cruz
NCT04809311	Remote Study Collecting Blood Glucose Values and Activity Da...	COMPLETED	NA	Novo Nordisk A/S
NCT00159575	Metformin in Assisted Reproduction-MET-AR-study	TERMINATED	PHASE4	Norwegian University of Science and Technology
NCT05981365	Voxelotor CYP and Transporter Cocktail Interaction Study	COMPLETED	PHASE1	Pfizer
NCT00374907	A Study Assessing Saxagliptin Treatment in Subjects With Typ...	COMPLETED	PHASE3	AstraZeneca

4. MARKET ANALYSIS & SALES INTELLIGENCE

Market Insights & Competitive Intensity:

Commercial sales records identified in 4 key market region(s). Primary market valuation stands at \$5,120.8M USD. Overall Market Feasibility Sub-Score: 90.0/100.

Region	Therapeutic Area	Year	Market Size (\$M)	CAGR %	Competitors
Global	Diabetes - Type 2	2023	\$5,120.8M	6.2%	48
United States	Diabetes - Type 2	2023	\$1,940.5M	4.8%	25
Europe	Diabetes - Type 2	2023	\$1,310.0M	5.1%	19
Global	Diabetes - Type 2	2022	\$4,820.5M	6.2%	45
Asia Pacific	Diabetes - Type 2	2022	\$1,420.6M	8.9%	31
United States	Diabetes - Type 2	2022	\$1,850.0M	4.8%	23
Europe	Diabetes - Type 2	2022	\$1,240.3M	5.1%	18

5. PATENT LANDSCAPE & FREEDOM-TO-OPERATE

Patent Landscape & Freedom-To-Operate (FTO):

Freedom-To-Operate Assessment: 'At Risk — 3 active patent constraint(s)'. Identified 7 patent filing(s), including 3 active patent(s) in force and 3 active constraint(s). Patent Landscape Sub-Score: 30.0/100.

Patent No.	Jurisdiction	Status	Expiry Date	Assignee	FTO Status
IN202118044291	IN	Active	2041-09-30	CSIR India	At Risk
CN112245409	CN	Pending	2040-11-25	Fosun Pharma	Free to Operate
JP2022512345	JP	Active	2039-03-15	Takeda Pharmaceutical	At Risk

US8846100	US	Active	2030-06-14	Sun Pharmaceutical Industries	At Risk
EP0567512	EU	Expired	2012-03-10	Bristol-Myers Squibb	Free to Operate
GB814978	GB	Expired	1976-04-02	Aron S.A. (Historical)	Free to Operate
US2868740	US	Expired	1972-05-20	Aron S.A. (Historical)	Free to Operate

6. SCIENTIFIC MOMENTUM & PUBLICATIONS

Scientific Momentum & Academic Evidence:

Identified 128,952 publication(s) in PubMed / Europe PMC repository. Includes 10 high-impact highly cited paper(s) (>=10 citations). Scientific Research Sub-Score: 100.0/100.

Publication Title	Journal	Year	Citations
Understanding the Warburg effect: the metabolic requirements of c...	Science (New York, N.Y.)	2009	12,125
Reduction in the incidence of type 2 diabetes with lifestyle inte...	The New England journal of medicine	2002	11,963
The hallmarks of aging	Cell	2013	11,617
Global, regional, and national incidence, prevalence, and years l...	Lancet (London, England)	2018	9,612
Dementia prevention, intervention, and care: 2020 report of the L...	Lancet (London, England)	2020	7,058
mTOR signaling in growth control and disease	Cell	2012	6,836

7. STRATEGIC RISK FACTORS

Strategic Risk Factors:

- Intellectual Property Risk: 3 active patent constraint(s) may present FTO blocking risks.

8. STRATEGIC INVESTMENT RECOMMENDATION

Investment Recommendation: **HIGH OPPORTUNITY** (Score: 75.8/100). Proceed with advanced commercial planning and licensing formulation.

9. APPENDIX & SYSTEM METADATA

Metadata Metric	Pipeline Value
Target Molecule Name	Metformin
Pipeline Version	1.0.0
Timestamp (UTC)	2026-07-23T18:23:50.710770+00:00
Report Provenance / Model	MoleculeIQ C-Suite Executive Synthesizer (Fallback)

Available Research Domains	clinical, literature, market, patent
Total Clinical Trials Evaluated	10
Total Publications Scanned	128,952
Geographic Sales Regions	Global, United States, Europe, Asia Pacific
FTO Assessment Status	At Risk — 3 active patent constraint(s)